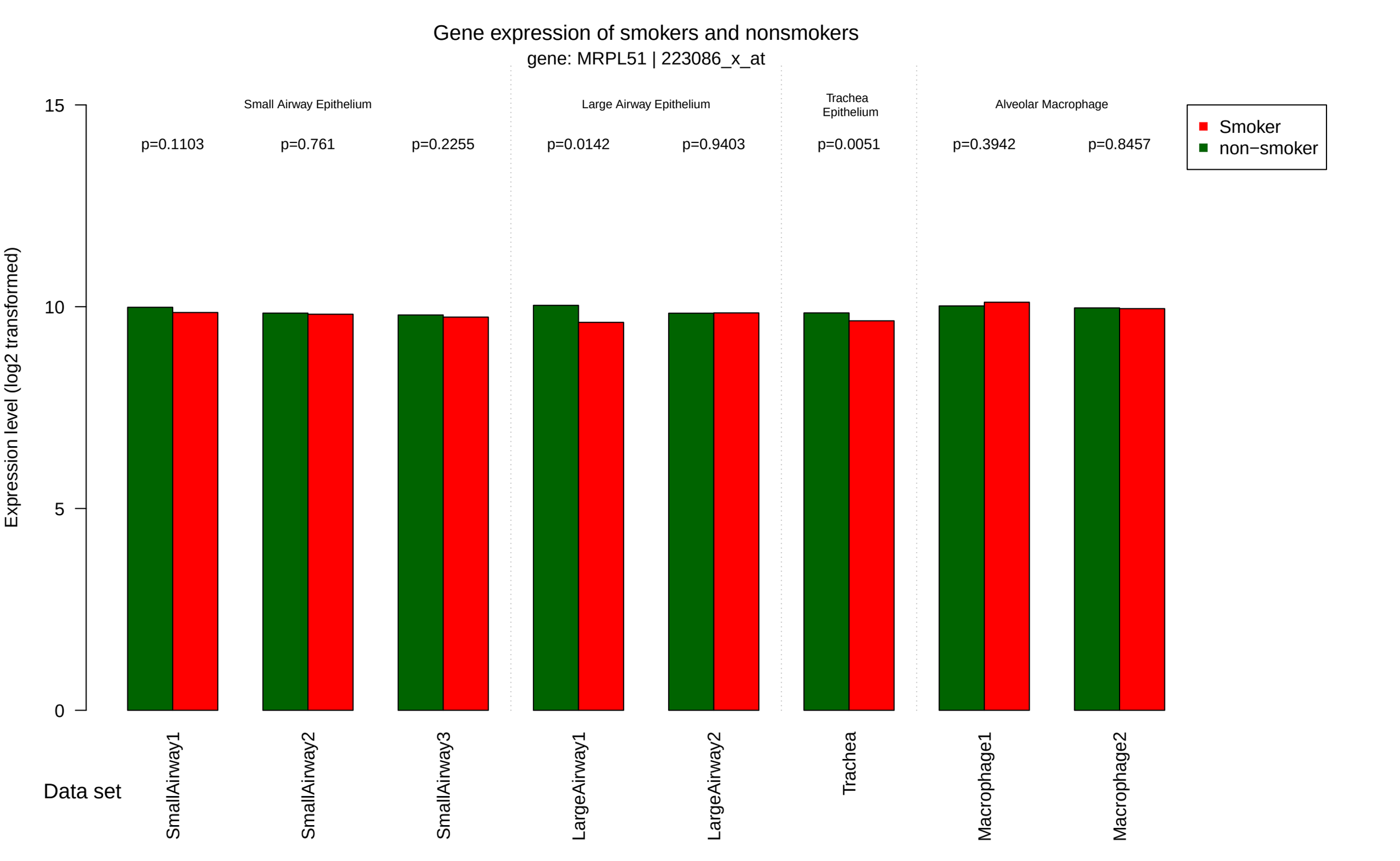
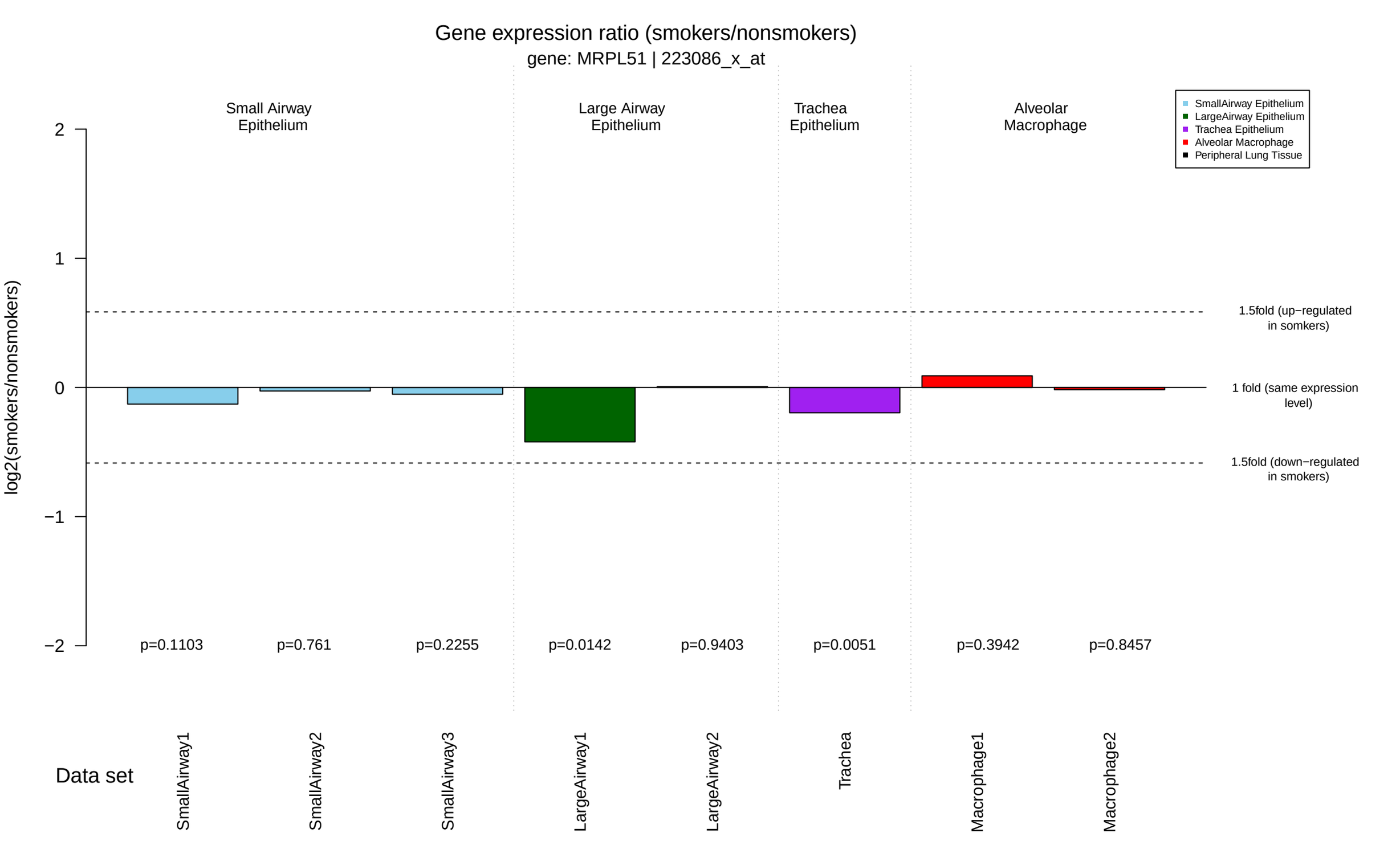
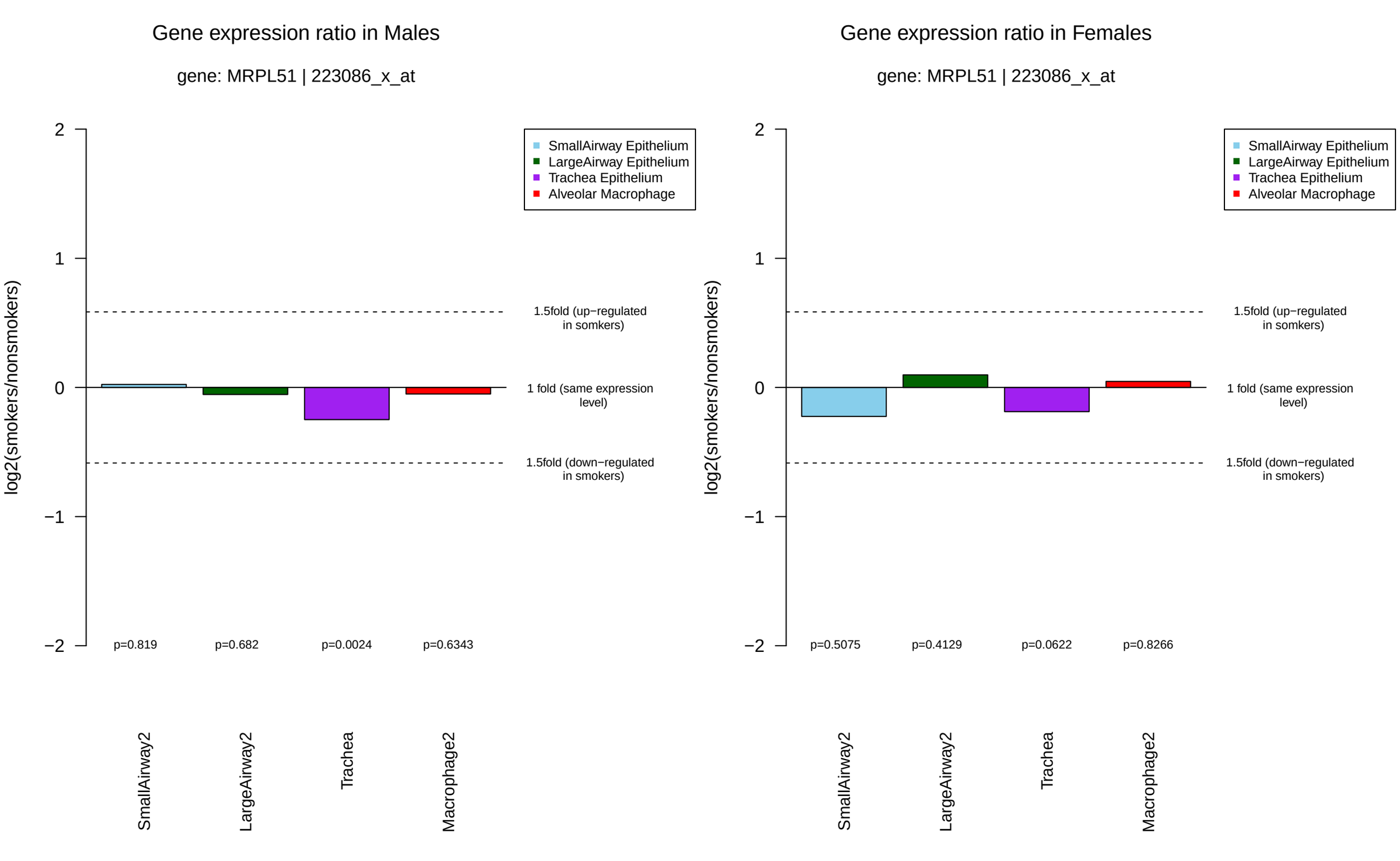
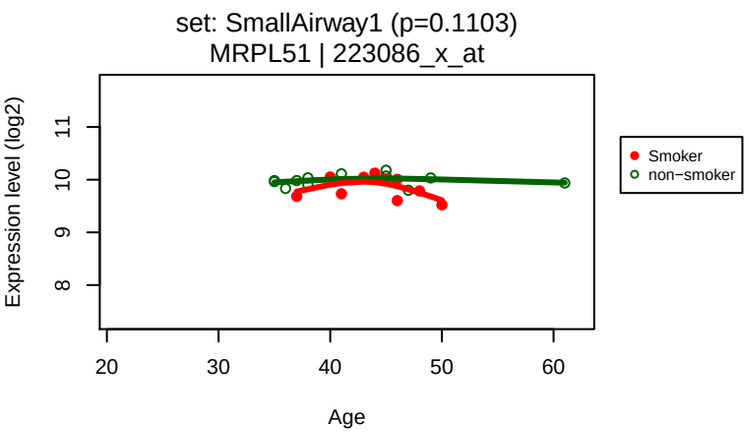
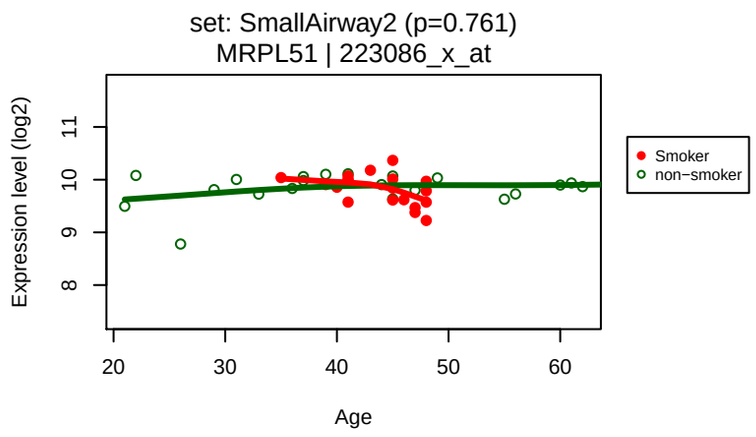
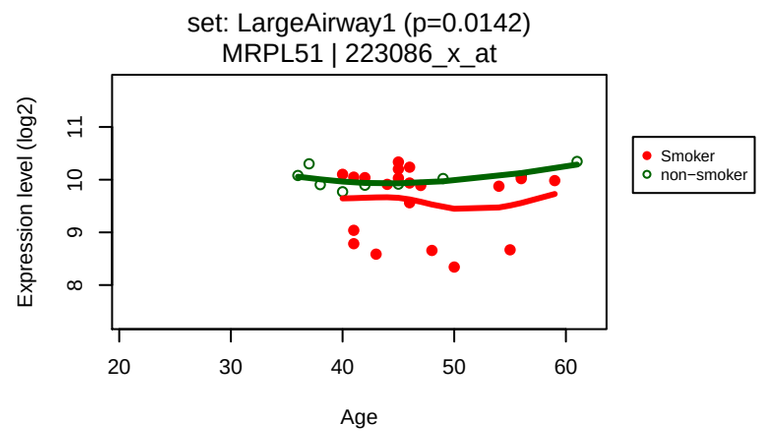

Gene expression of smokers and nonsmokers
gene: MRPL51 | 223086_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1103
p=0.761
p=0.2255
p=0.0142
p=0.9403
p=0.0051
p=0.3942
p=0.8457
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

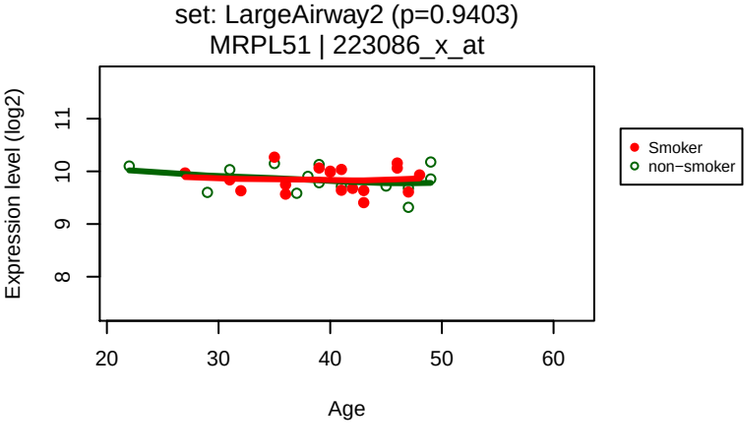

Gene expression ratio (smokers/nonsmokers)
gene: MRPL51 | 223086_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1103
p=0.761
p=0.2255
p=0.0142
p=0.9403
p=0.0051
p=0.3942
p=0.8457
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

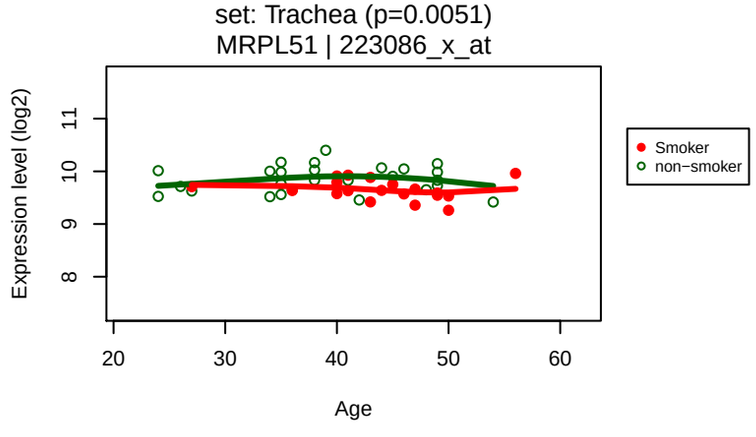

Gene expression ratio in Males
gene: MRPL51 | 223086_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.819
p=0.682
p=0.0024
p=0.6343
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MRPL51 | 223086_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5075
p=0.4129
p=0.0622
p=0.8266
SmallAirway2
LargeAirway2
Trachea
Macrophage2

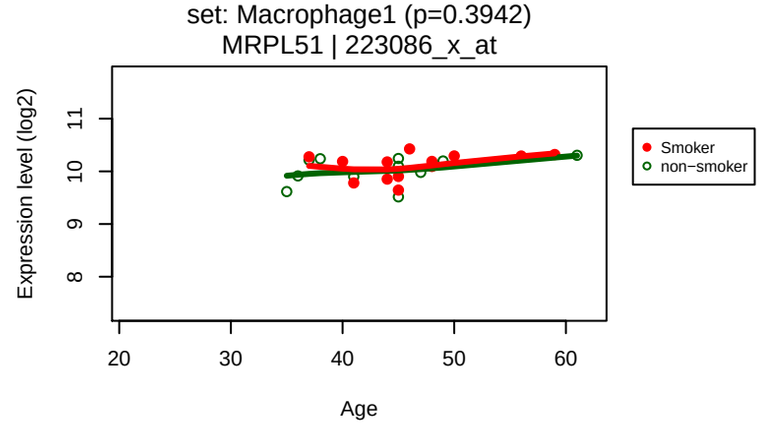

set: SmallAirway1 (p=0.1103)
MRPL51 | 223086_x_at
Expression level (log2)
Age
Smoker
non−smoker

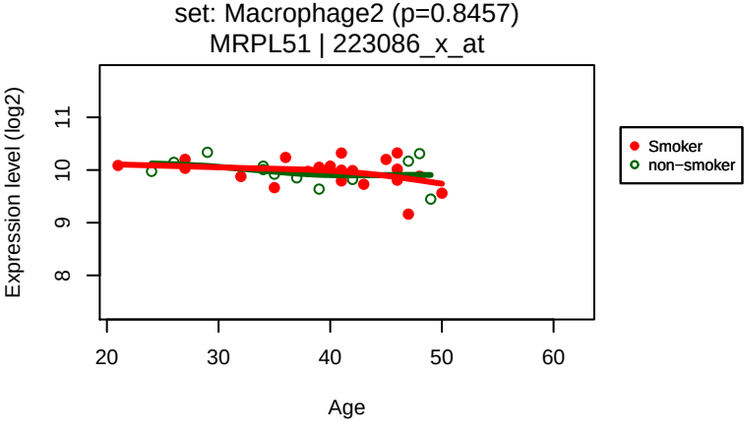

set: SmallAirway2 (p=0.761)
MRPL51 | 223086_x_at
Expression level (log2)
Age
Smoker
non−smoker

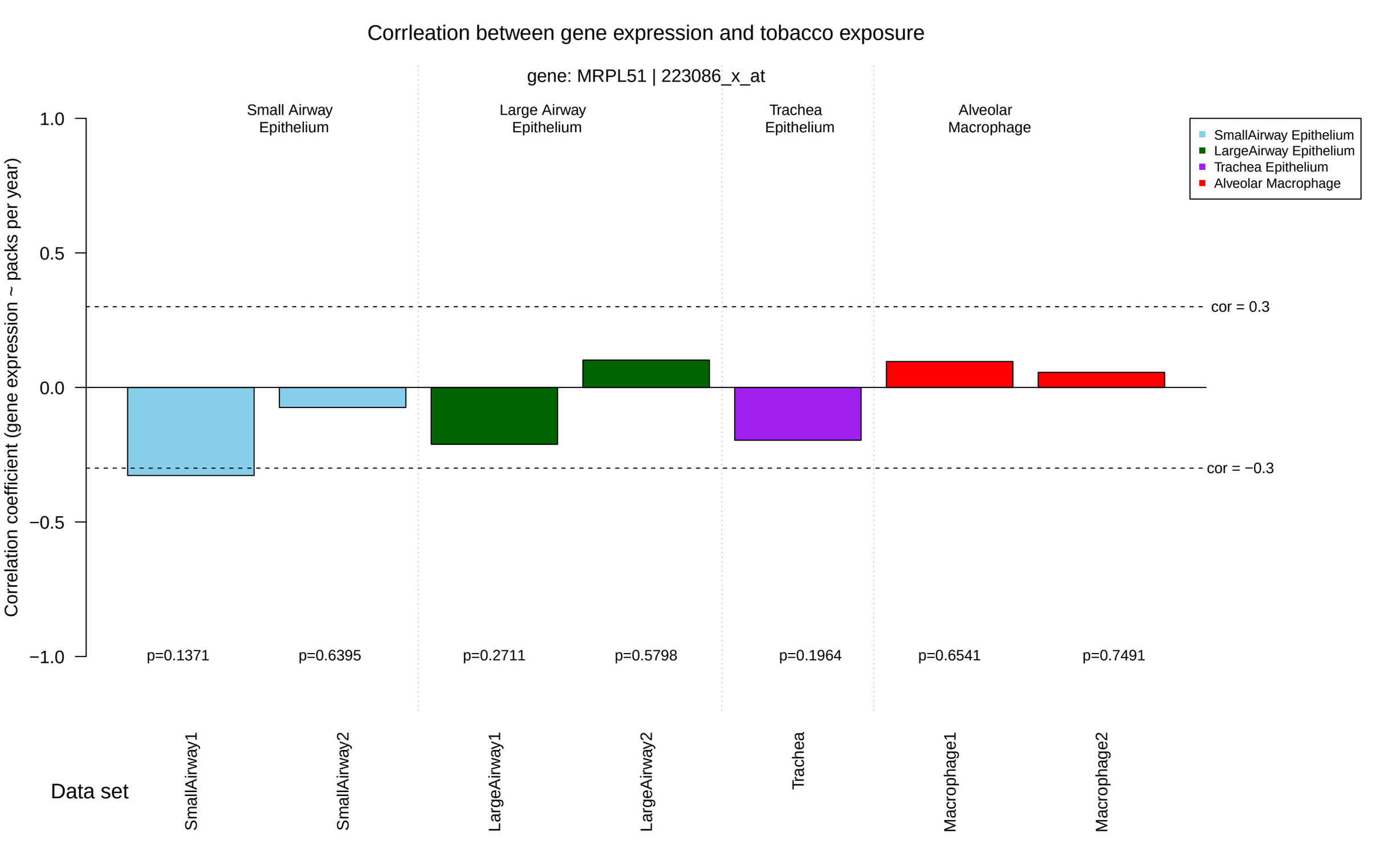

set: LargeAirway1 (p=0.0142)
MRPL51 | 223086_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9403)
MRPL51 | 223086_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0051)
MRPL51 | 223086_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3942)
MRPL51 | 223086_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8457)
MRPL51 | 223086_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MRPL51 | 223086_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1371
p=0.6395
p=0.2711
p=0.5798
p=0.1964
p=0.6541
p=0.7491
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2